Test Methodology and Practice for Multi-Accelerator Copper Scaleup Domain Interconnect and Communication Library

Yao Rong *Systems Engineer, AI Systems Meta*

Karthik Ganesan *AI/HPC System Performance Engineer, Meta*

OCP PLATINUM Member

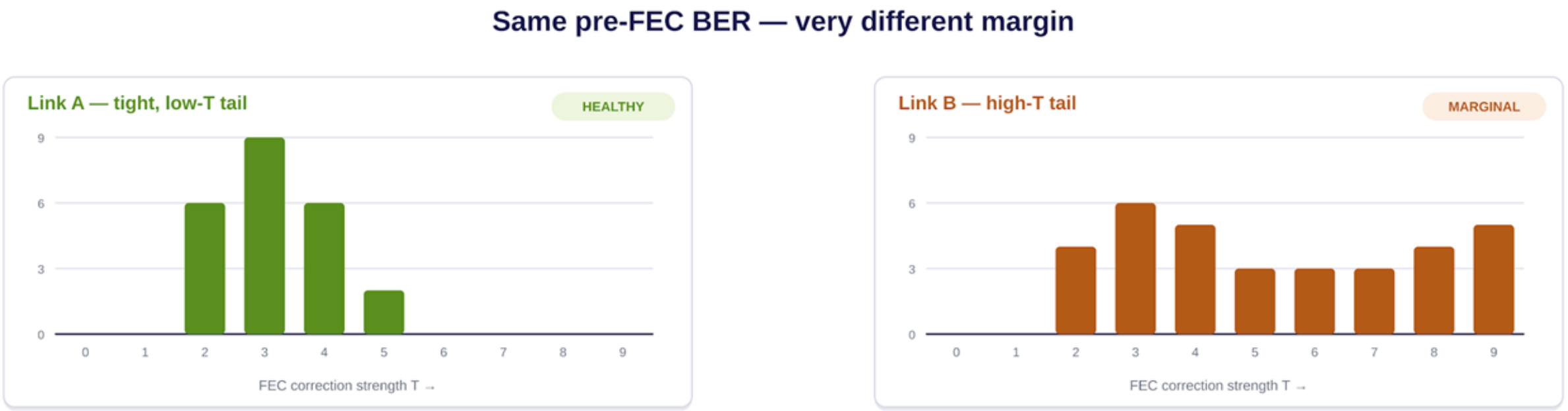
What a Copper Scale-Up Domain Is

- **A statistical problem, not a debug problem**
- Typical system: 72 accelerators · single switch hop
- Compute + switch trays + blind-mate cartridges
- Thousands of connections/rack → ~10 bad links even at 99.9%
- No per-link serviceability — the mate is now the signal path



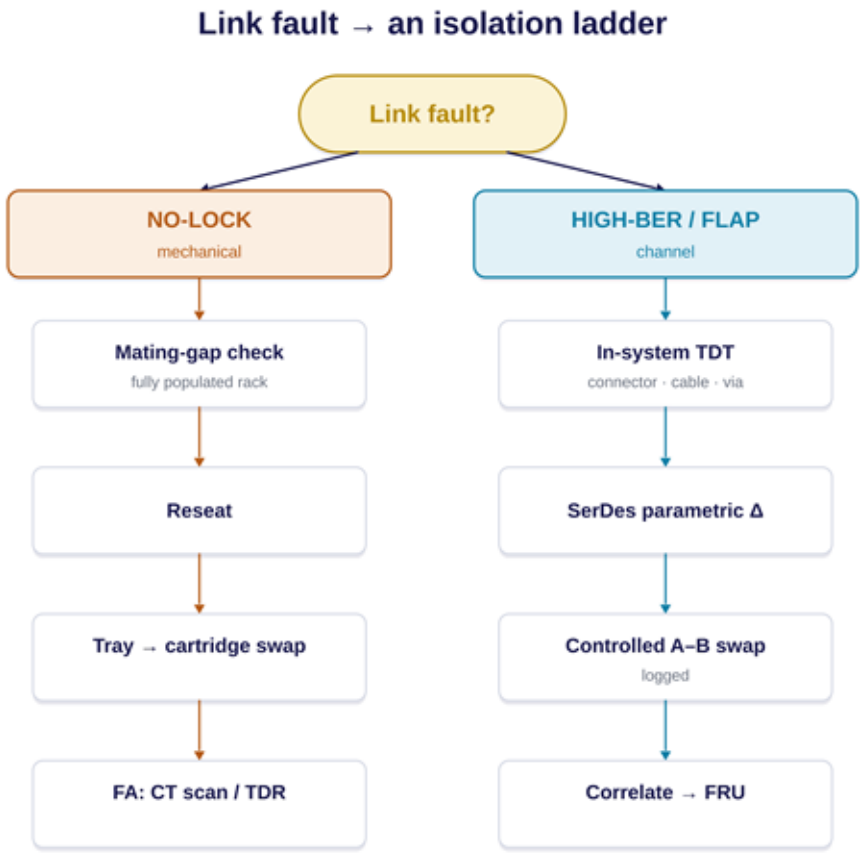
EE/SI Validation + High-Speed SerDes

- **Exit number + per-lane vector, not pass/fail**
- Ladder POC → EVT → DVT → PVT, each with an exit number
- Both pre-FEC and post-FEC error rate matter
- FEC histogram beats BER — same BER, different margin



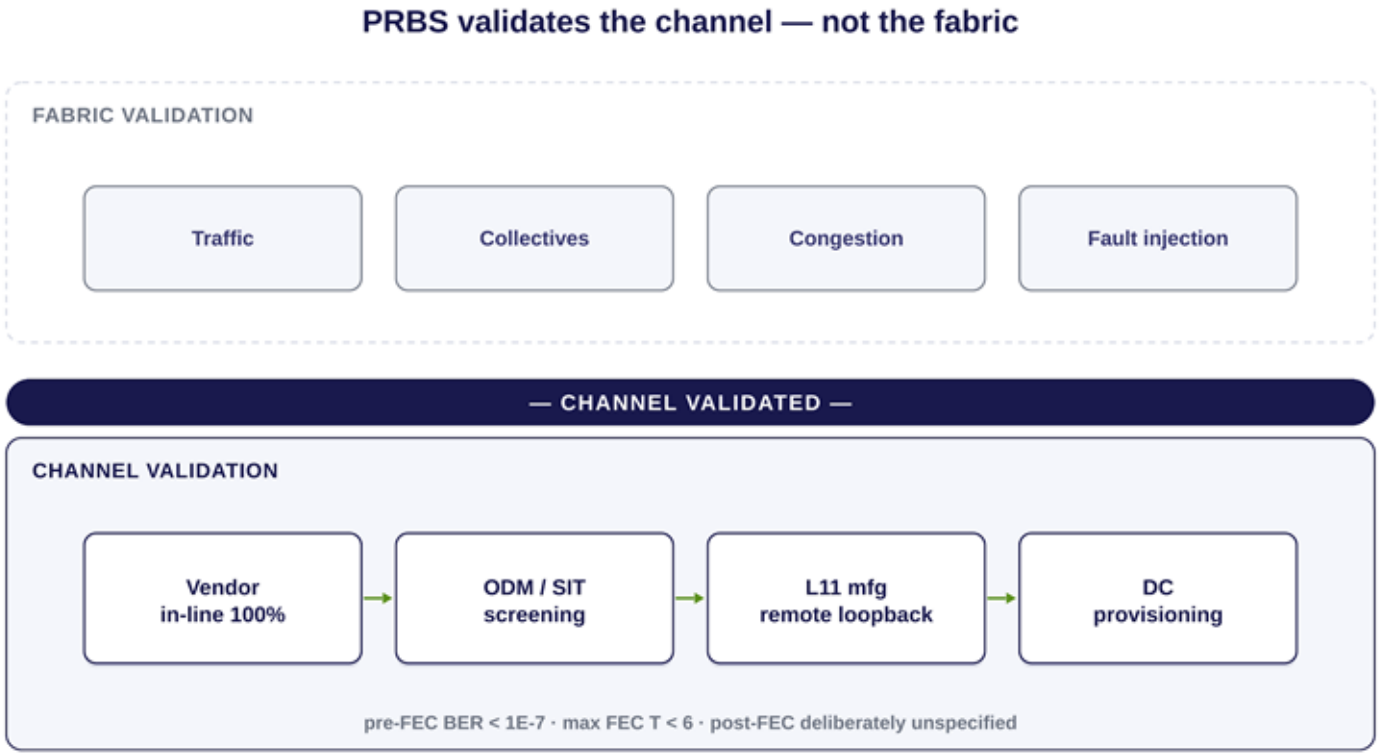
Link Down & Link Flap: An Isolation Ladder

- **"Reseated it and it got better" is not a root cause**
- Two classes: link down vs link flapping
- ME: Measure the de-mate gap on a fully populated rack
- Flapping is worse in debug than down → it becomes a straggler
- Detect on flap duration & rate is key



PRBS: What It Proves, and What It Cannot

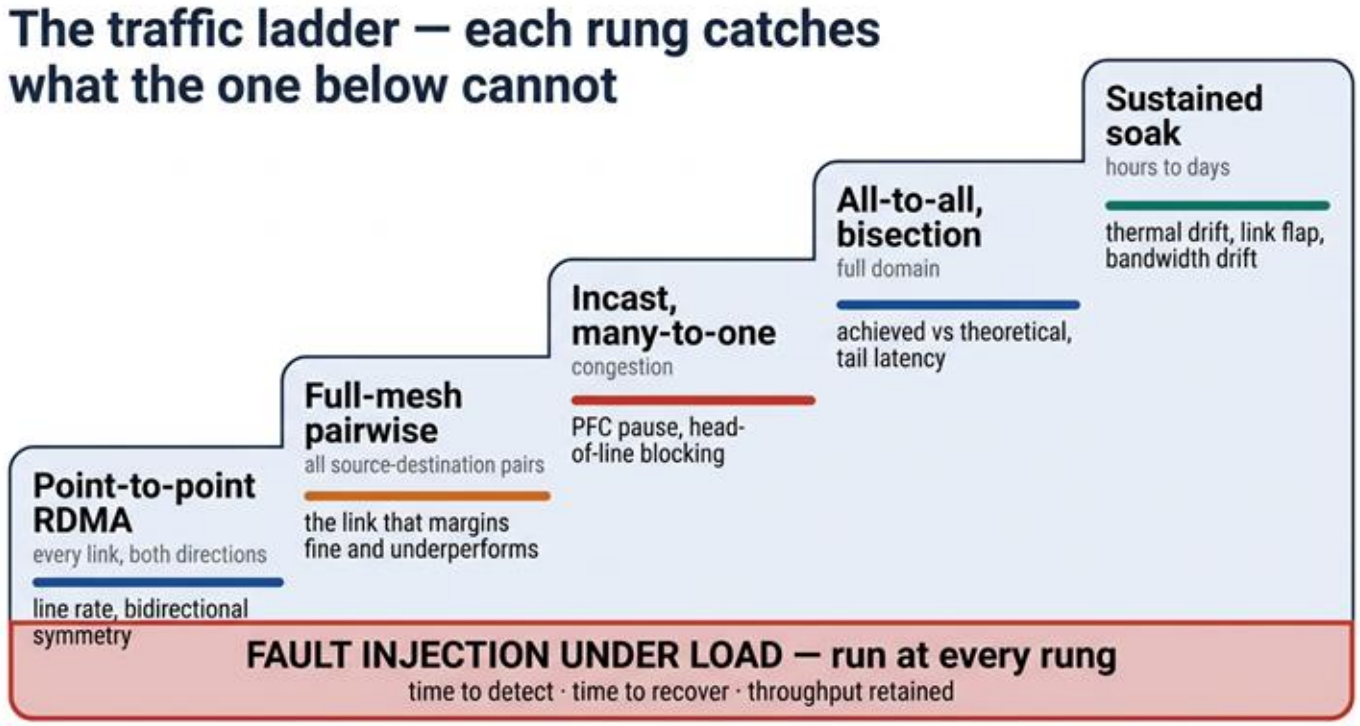
- **Necessary, not sufficient — the ladder continues**
- **Even if PRBS passes with low BER, PI and cross at runtime can still induce erros**
- SI data telemetry at scale is crucial.
- Fault isolation tool is important to avoid iterative repair.



Beyond PRBS: Traffic-level Validation

A clean link is not a working fabric

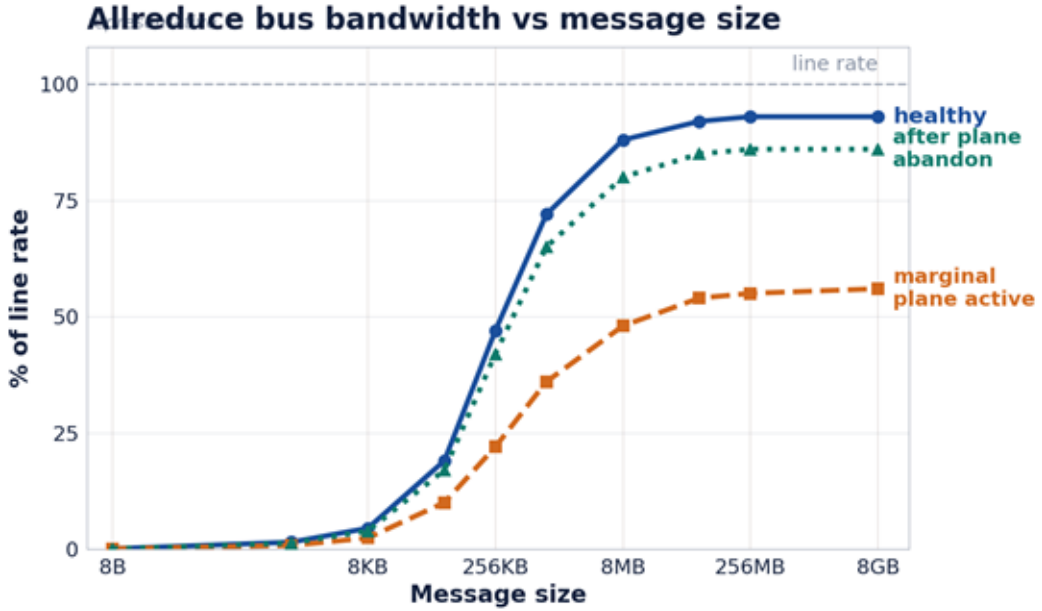
- PRBS validates the electrical channel
  - But, retransmission, congestion, or thermal drift under sustained load?
- Each rung catches what the one below cannot.
- The rung with no agreed pass/fail criteria: fault injection under load.
  - time to detect, recover or b/w retained
  - The reusable artifact is a vendor-neutral conformance table, not the numbers



Collectives are the Acceptance Test

One marginal link - worse than a dead link

- AllReduce (ring) performance - slowest link
- Dead link: fabric abandons plane -> Clean.
- Marginal link: plane stays active
- Go-Back-N makes it a straggler
- Test matrix:
  - ops x message size x scale x algorithm.
- Read scaling efficiency and the algorithm



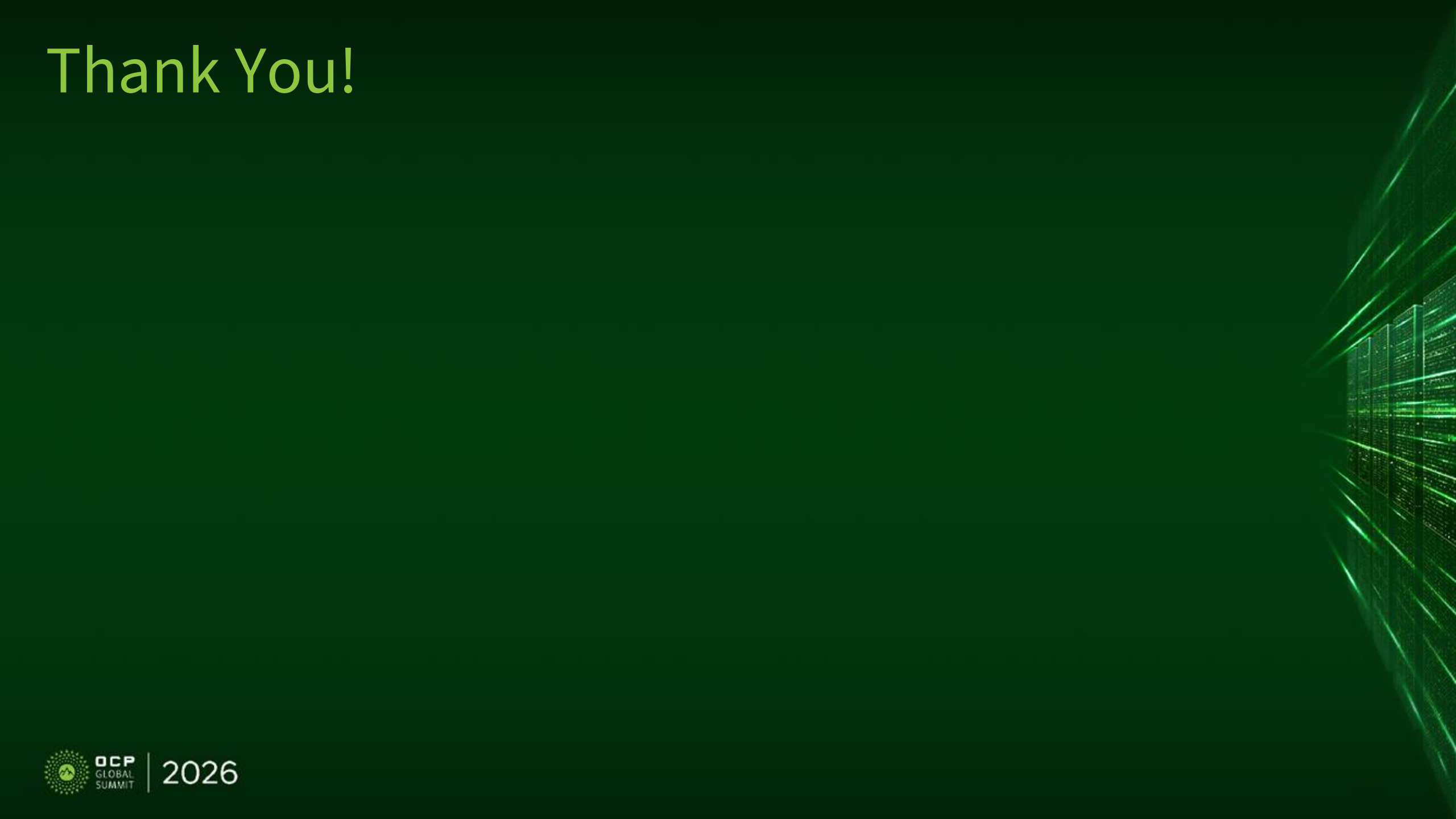


Call to Action

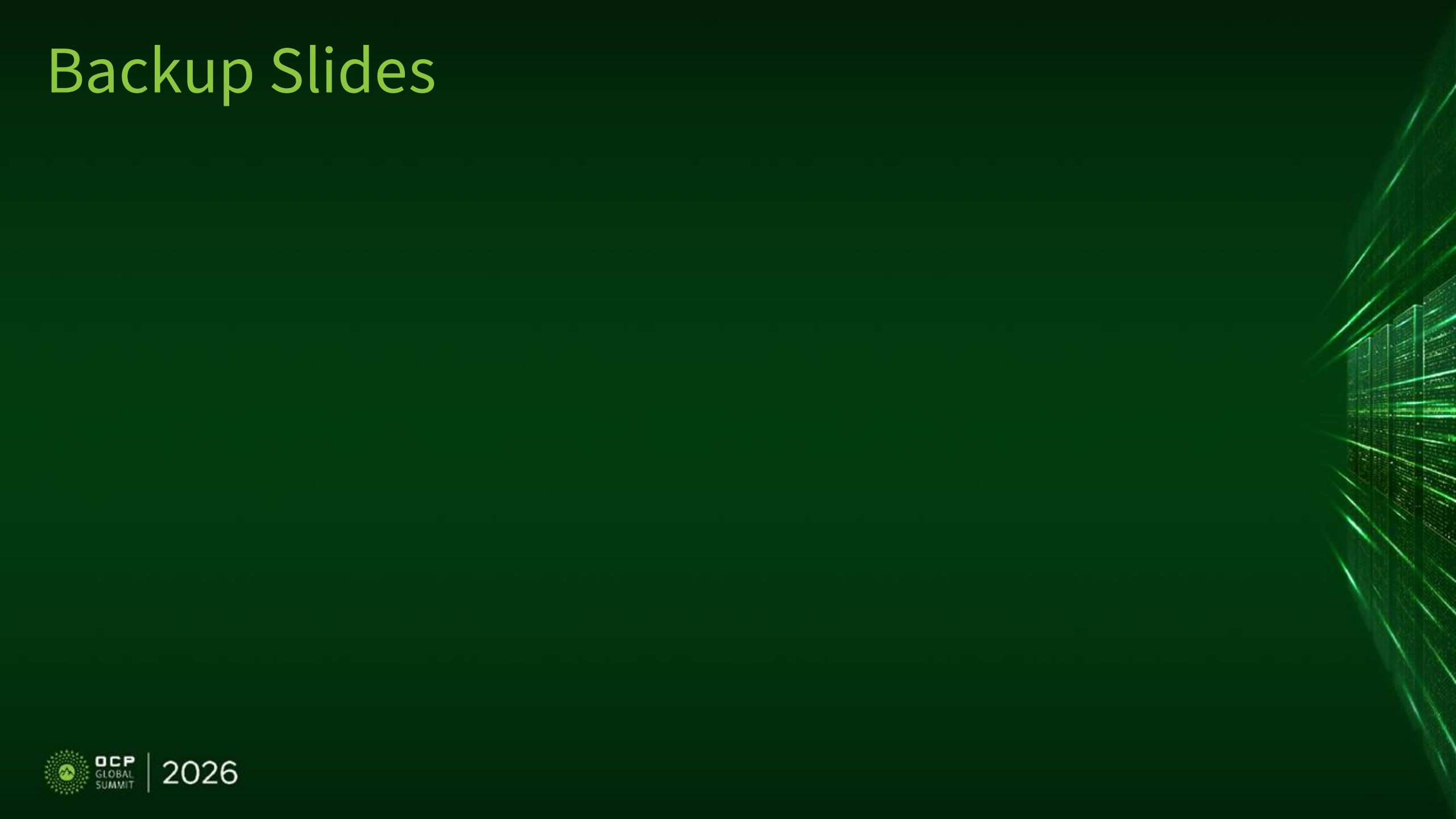
- A copper scale-up conformance suite.
  - The SI ladder, the traffic ladder, and the collective matrix, with defined metrics and pass/fail thresholds.
  - Benchmark definitions precise enough that results are comparable across accelerator SKUs and vendors, not merely reproducible inside one vendor's lab.
- Unified specification between different accelerator solution
- Detect/Flag Scale-up fabric performance differences at CCL level
  - E.g., significant variation across steps of an all-reduce
- Enhancement in diagnostics of passive component failure in the field will yield great technical and economic benefits

"Every vendor is building a copper scale-up domain. Every one of us is inventing our own acceptance test."

Thank You!



Backup Slides



Case study: PRBS-clean, traffic-degraded

SYMPTOM: Collective throughput drops across full domain [to be populated]

ISOLATION: Full-mesh pairwise heatmap shows one GPU degraded across an entire plane - one marginal link drags all GPUs in the domain.

ROOT CAUSE: [Physical finding, genericized - to be populated]

FIX: Plane abandoned, traffic redistributed, bandwidth restored.

